

GSSSP : Generator of Synthetic Scans of Spectroscopic Plates for Object Detection

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Abstract. GSSSP (Generator of Synthetic Scans of Spectroscopic Plates) is a tool for generating labeled scans of physical spectroscopic observations. The area of spectroscopic analysis is old in the Astronomic Domain. Legacy observational methods recorded stellar spectra on photographic plates. These plates have high historical and scientific value. However, a limited number of them have been digitized, resulting in an insufficient amount of data to train modern computer vision models for observation detection.

In this work, we present a software tool that replicates the appearance of scanned spectroscopic plates by generating synthetic images containing spectra of artificial stellar objects. The generated images are accompanied by label files that enable the training of observation detection models.

We trained a detection model and compared its performance when using only real data, only generated data, and a combination of both. The results show that GSSSP enables the model to learn the general visual patterns of this type of plate, providing a strong generalization basis for subsequent fine-tuning. The resulting model surpasses the current state of the art for this problem.

Keywords: Generated data, Object detection, Spectroscopic plates

1 Spectroscopic Data

Spectroscopy constitutes a fundamental tool for physic study of astronomical objects. The analysis of electromagnetic radiation received, absorbed, or reflected is possible infer physics propieties of objects. By example, quimic composition, temperature, radial speed. This science enable celest bodis study further of her position or apparent brightness [?].

In XX century the capture of spectroscopic data was very popular. The method consist in capture spectroscopic data through photography. With the peculiarity what the image persisted in glass plates instead of paper. Today many of these plates are conserved for their institutions [?]. However, the persistence of data in physical plates can present problems under suboptimal storage conditions. Figure 1 show a spectroscopic glass plate with two observations in its surface.

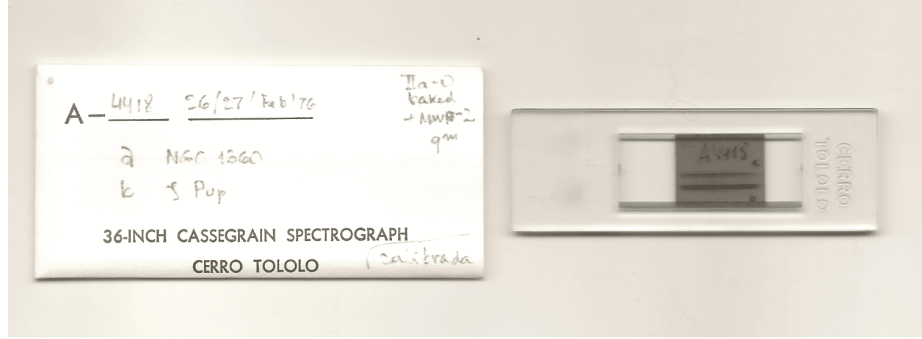


Fig. 1: Right, spectroscopic plate A4418 of astronomical source Zeta Puppis captured at Observatorio Interamericano de Cerro Tololo (Chile) as part of the ReTrOH [?] project. The plate contains two spectroscopic observations. Left, wrapping, includes additional information about each observation.

Accidents have occurred in involving this type of heritage. In 2016, due a burst pipe, Harvard College Observatory suffered a flood that affected more than 61.000 plates in its colection [?]. Too, Facultad de Ciencias Astronomicas y Geofisicas de la Universidad Nacional de La Plata found an equally damaging situation, discovering mold on the furniture where the plaques were stored [?]. The risk of mold stains, scratches, cobwebs, or similar problems is significant and, if not properly addressed, can lead to consequences that permanently degrade the information available on each plate.

In addition to physic deterioration can be introduced defects exist defects related to the capture process such as curvatures in the trace of the spectra, artifacts, or the loss of certain elements. Figure 2 show refences of plates with some defect.

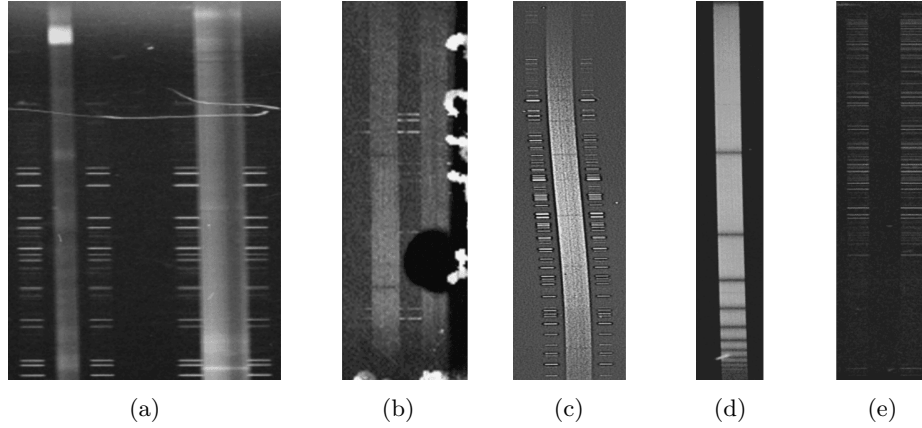


Fig. 2: Samples of possible defects on the plates. Defects associated with prolonged storage: (a) Scratches. (b) Dirt and/or moisture stains. Defects associated with the capture process: (c) Curved engraving. (d) Missing comparison lamps. (e) Comparison lamps without their respective spectrum.

For preserve this heritage and institutions as IAU (International Astronomical Union) presented resolutions⁵. when remark the importance of keep spectroscopic plates for next generations of astronomers. IAU recommends digitalization as conservation method. This methods enable copying facilities and redundancies that can preserve the most relevant information in the long term.

For advance in this many institutions begin projects of digitalization of their collections. Table 1 list many collections with big count of spectroscopic collections. Theirs digitalization percents and his calibration percents.

Colección	#plt.fot.	#plt.spe.	Digitalizado	Calibrado	Rango
HCO [?]	~ 430.000	0	100%	89%	1880–1992
APPLAUSE [?]	93.774	19.240	100%	83% ^(a)	1893–1999
CPDP [?]	30.750	0	95%	51%	1901–1999
MMO [?,?]	>8.500	0	94%	0%	1913–1993

continúa en la página siguiente

⁵ UAI Resolution B3 (Agosto 2000): Safeguarding the Information in Photographic Observations

Colección	#plt.fot.	#plt.spe.	Digitalizado	Calibrado	Rango
ReTrHO [?]	?	>15.000	4%	0%	1920–1980
WFPDB [?,?]	640.000	?	100%	?%	1858–2005
OCA Archive [?]	11.800	0	100%	?%	1935–1996
OHP Archive [?]	7.300	63.500	100%	?%	1944–1997
Pulkovo Archive ^(b) [?]	46.483	0	100%	?%	1893–2005
Foster Archive [?]	0	>4.800	100%	?%	1928–1991
HDAP [?,?]	~25.000	0	100%	?%	1880–1999
DFBS [?]	0	1.874	100%	100%	1965–1980
AO KNU ^(c) [?]	20.000	0	22%	0%	1898–1996
AO LNU ^(d) [?]	8.000	0	25%	0%	1939–1976

Table 1: Varios proyectos y colecciones relevantes de placas fotográficas y espectroscópicas.

^a Solo se calibraron las placas fotográficas.

^b *Pulkovo Photographic Plate Database (Noviembre 2025)*. Disponible en <https://www.puldb.ru/db/plates/>.

^c *Ukrainian Virtual Observatory, AO KNU DA (Noviembre 2025)*. Disponible en <http://ukr-vo.org/digarchives/index.php?b4&1>.

^d *Ukrainian Virtual Observatory, AO LNU DA (Noviembre 2025)*. Disponible en <http://ukr-vo.org/digarchives/index.php?b3&1>.

Descripción de las placas, relevancia, contexto de que hay pocas.

1.1 Available Data

For this study, we use a dataset composed of 361 spectroscopic plates from **Cerro Tololo** that were scanned and labeled for observation detection. The dataset is

publicly available ⁶. Due to the limited availability of digitized spectroscopic plates, the dataset size is relatively small.

2 State of Art

Mencion a trabajos previos donde se construyeron modelos de deteccion para observaciones.

3 GSSSP

En que consiste GSSSP, sus modulos. Caracteristicas principales, personalizacion de observaciones e imagenes de espectros.

4 Experiment

Detalles del entrenamiento, datos usados en cada uno y metricas de evaluacion.

4.1 Results

Resultados de cada entrenamiento.

Table 2: Results with real, generated and mix data train.

data train	data test	MaP	$MaP[IoU = 50]$	$MaP[IoU = 75]$
<i>real-train</i>	<i>real-test</i>	0.6597	0.9114	0.6681
<i>gen</i>	<i>real-test</i>	0.6431	0.8942	0.7675
<i>gen+real-train</i>	<i>real-test</i>	0.9360	1.0000	1.0000

5 Conclusion

Conclusions.

6 Sample Section

6.1 A Subsection Sample

Please note that the first paragraph of a section or subsection is not indented. The first paragraph that follows a table, figure, equation etc. does not need an indent, either.

Subsequent paragraphs, however, are indented.

⁶ Observation Detection in Spectroscopic Plates Dataset

Sample Heading (Third Level) Only two levels of headings should be numbered. Lower level headings remain unnumbered; they are formatted as run-in headings.

Sample Heading (Fourth Level) The contribution should contain no more than four levels of headings. Table 3 gives a summary of all heading levels.

Table 3: Table captions should be placed above the tables.

Heading level	Example	Font size and style
Title (centered)	Lecture Notes	14 point, bold
1st-level heading	1 Introduction	12 point, bold
2nd-level heading	2.1 Printing Area	10 point, bold
3rd-level heading	Run-in Heading in Bold. Text follows	10 point, bold
4th-level heading	<i>Lowest Level Heading.</i> Text follows	10 point, italic

Displayed equations are centered and set on a separate line.

$$x + y = z \tag{1}$$

Please try to avoid rasterized images for line-art diagrams and schemas. Whenever possible, use vector graphics instead (see Fig. 3).

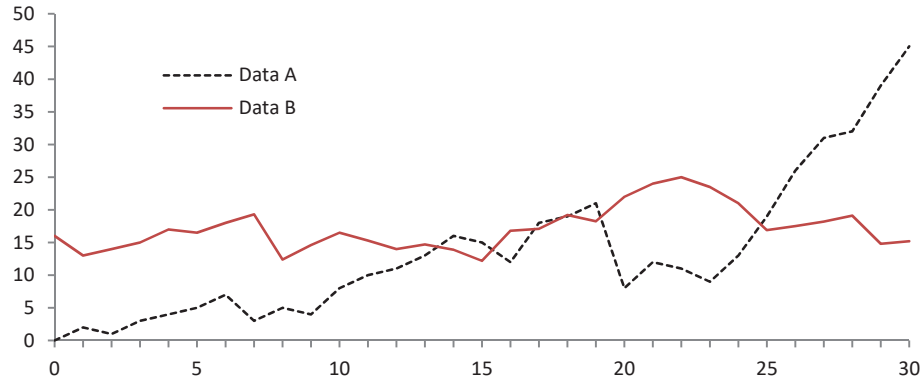


Fig. 3: A figure caption is always placed below the illustration. Please note that short captions are centered, while long ones are justified by the macro package automatically.

Theorem 1. *This is a sample theorem. The run-in heading is set in bold, while the following text appears in italics. Definitions, lemmas, propositions, and corollaries are styled the same way.*

Proof. Proofs, examples, and remarks have the initial word in italics, while the following text appears in normal font.

For citations of references, we prefer the use of square brackets and consecutive numbers. Citations using labels or the author/year convention are also acceptable. The following bibliography provides a sample reference list with entries for journal articles [1], an LNCS chapter [2], a book [3], proceedings without editors [4], and a homepage [5]. Multiple citations are grouped [1,2,3], [1,3,4,5].

References

1. Author, F.: Article title. Journal **2**(5), 99–110 (2016)
2. Author, F., Author, S.: Title of a proceedings paper. In: Editor, F., Editor, S. (eds.) CONFERENCE 2016, LNCS, vol. 9999, pp. 1–13. Springer, Heidelberg (2016). <https://doi.org/10.1007/1234567890>
3. Author, F., Author, S., Author, T.: Book title. 2nd edn. Publisher, Location (1999)
4. Author, A.-B.: Contribution title. In: 9th International Proceedings on Proceedings, pp. 1–2. Publisher, Location (2010)
5. LNCS Homepage, <http://www.springer.com/lncs>. Last accessed 4 Oct 2017